Practice Set 5 Paper 4 – Recent Advances

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Practice paper compiled from the question bank.

Q1. Describe energy sources for braking stones in uro

Type: LAQ (25 marks) (also answered in Practice Set 1 Paper 4, Q2)

Read as "energy sources for breaking stones in urology".

Introduction

- Stones are **brittle**: they crack under tension and shear. Tissue is **elastic** and survives the same stress. Every lithotripter exploits this difference.
- Energy reaches the stone **extracorporeally** (SWL, shock wave lithotripsy) or **intracorporeally** through an endoscope.

Classification

Group	Sources
Extracorporeal	SWL: electrohydraulic, electromagnetic, piezoelectric; BWL (burst wave lithotripsy, investigational)
Intracorporeal mechanical	Ballistic (pneumatic), ultrasonic, combined ultrasonic–pneumatic
Intracorporeal electrical	EHL (electrohydraulic lithotripsy)
Intracorporeal light	Lasers: Ho:YAG (holmium:yttrium-aluminium-garnet), TFL (thulium fibre laser), pulsed Tm:YAG (thulium:YAG); historic pulsed dye and FREDDY

1. Shock wave lithotripsy

- **Shock wave:** peak positive pressure 30–100 MPa, then a tensile tail of about −10 MPa. It is generated at F1, focused, and delivered to the stone at F2.

Generator	Mechanism	Strength	Weakness
Electrohydraulic (original HM3, 1980)	Underwater spark at F1 of an ellipsoid reflector	Very effective	Electrode life <40,000 shocks; focus drifts
Electromagnetic (modern standard)	Coil pushes a metal membrane; lens or reflector focuses	Consistent; high energy at F2; long life	—
Piezoelectric	Many ceramic elements on a spherical dish	Long life; little pain	Small focal zone, low energy

- **Fracture mechanisms:** shear and tear forces; **spallation** at the distal face; quasistatic squeezing; **cavitation** (bubble collapse erodes the surface); dynamic squeezing (most powerful). A focal zone wider than the stone works best.
- **Technique:** 60–90 shocks/min beats 120/min, because bubble clouds shield the stone and injure the kidney; voltage ramping; bubble-free gel coupling.
- Poor results: calcium oxalate monohydrate, brushite, cystine, density >1000 HU (Hounsfield units), skin-to-stone distance >10 cm.

2. Ballistic (pneumatic) lithotripsy

- Compressed air drives a projectile against a metal probe, a **jackhammer** (about 12 Hz).
- Breaks all stones cheaply with little ureteric injury. Drawbacks: **retropulsion**; rigid probes only, as flexible probes block deflection.

3. Ultrasonic lithotripsy

- A piezoceramic transducer vibrates a hollow probe at ultrasonic frequency; contact drills the stone.
- **Suction** through the hollow probe removes fragments at once. Best for soft stones in PCNL (percutaneous nephrolithotomy); weak on hard stones; needs irrigation for cooling.

4. Combined devices

- Ultrasonic plus ballistic in one probe (for example Trilogy, ShockPulse-SE): ballistic cracks hard stones while ultrasound and suction clear fragments. Faster PCNL.

5. Electrohydraulic lithotripsy

- A spark at the electrode tip forms a **vapour bubble**; expansion and collapse send pressure waves in all directions.
- Cheap; 3–5 Fr flexible probes. Poorly focused, so it injures the ureter and the scope; weak on cystine. AUA (American Urological Association) advises against it as first-line ureteric lithotripsy; still used for bladder stones.

6. Lasers

- **Pulse duration decides mechanism.** Short pulses (<10 µs: pulsed dye, FREDDY) are photomechanical, with cavitation shocks. Long pulses (>250 µs: holmium, thulium) are **photothermal**: the stone is heated until it melts, decomposes or flashes its water to steam.

	Ho:YAG	TFL
Wavelength	**2120 nm** (often quoted 2100)	1940 nm (water absorption peak)
Frequency	Up to about 120 Hz	Up to about 2000 Hz
Peak power	High	Low → less retropulsion
Fibre	≥200 µm	50–150 µm (more flow and deflection)
Product	Fragments and dust	Finer dust

- **Holmium:** standard since the 1990s; breaks **every** composition; flexible quartz fibres. Fragmenting 0.6–1.0 J × 6–10 Hz; dusting 0.2–0.3 J × 40–80 Hz; "popcorning" in a calyx.
- **Moses effect:** the start of the pulse boils a vapour channel through which the rest reaches the stone; pulse modulation reduces retropulsion.
- **TFL RCTs (randomised controlled trials):** superior to low-power (30 W) holmium; equal stone-free rate to 120 W pulse-modulated holmium.
- **Retropulsion** rises with pulse energy, not frequency, and falls with longer pulses.

Safety

- Laser heat follows total power: ≤10 W in the ureter, pauses, high flow, access sheath. Eyewear for all.
- Never fire without seeing the fibre tip on the stone.
- SWL: haematoma, arrhythmia, steinstrasse. Contraindications: pregnancy, coagulopathy or anticoagulants, untreated infection, aneurysm, distal obstruction.

Choice by site

Site	Usual energy
Ureter	Laser (semirigid or flexible); pneumatic distally; SWL
Kidney	SWL; laser via flexible URS (ureteroscopy); PCNL with ultrasonic, combined or laser
Bladder	Laser, pneumatic, EHL, mechanical

Recent advances

- **BWL:** handheld ultrasound bursts (5–100 cycles) break stones without anaesthesia. **Ultrasonic propulsion** moves fragments.
- Suction sheaths; AI (artificial intelligence)-adjusted laser settings.

Diagram to draw: SWL electrohydraulic ellipsoid with F1 and F2, stone at F2.

Key points to remember:

- SWL: electromagnetic is the modern standard; slow rate (60–90/min) is better.
- Holmium 2120 nm, photothermal, breaks every composition.
- TFL 1940 nm: low peak power, finer dust, less retropulsion.
- EHL is poorly focused, so it is not first-line in the ureter.

Q2. Discuss advances in management of CA prostate

Type: LAQ (25 marks) (related: 2025-Winter Paper 3, Q3b; 2024-Winter Paper 4, Q3f)

Introduction

- Three aims drive recent change: **find significant cancer and avoid diagnosing insignificant cancer**; **match local treatment to risk** with less morbidity; give **earlier, intensified and targeted** systemic therapy.

1. Screening and diagnosis

- **ERSPC (European Randomized Study of Screening for Prostate Cancer)** and Göteborg-1: PSA (prostate-specific antigen) screening lowers cancer-specific mortality, but PSA alone causes negative biopsies and overdiagnosis of GG (Grade Group) 1. Hence risk-stratified pathways.
- **Biomarkers** (reflex tests before biopsy):

Test	Sample	Content
PHI (Prostate Health Index)	Blood	(p2PSA ÷ free PSA) × √total PSA
4Kscore	Blood	Four kallikreins + clinical data; % risk of GG ≥2
STHLM3	Blood	4Kscore kallikreins + 232 risk SNPs (single-nucleotide polymorphisms) + clinical variables
PCA3 (prostate cancer antigen 3), SelectMDx	Urine after DRE (digital rectal examination)	RNA ratio (PCA3); DLX1 and HOXC6 (SelectMDx)
ExoDx IntelliScore	Urine, no DRE	Exosomal RNA of PCA3, ERG, SPDEF

- Tissue genomic classifiers (Decipher, Prolaris, Oncotype DX) refine risk for surveillance and adjuvant decisions.
- **MRI-first pathway:** mpMRI (multiparametric magnetic resonance imaging), scored by **PI-RADS (Prostate Imaging–Reporting and Data System)** 1–5, before biopsy.
  - **PROMIS (2017):** mpMRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy; NPV (negative predictive value) 89%; 27% could avoid biopsy.
  - **PRECISION (2018):** MRI with targeted biopsy of PI-RADS ≥3 found more significant cancer (38% vs 26%); 28% avoided biopsy.
  - So PI-RADS ≥3 lesions are biopsied, while PI-RADS 1–2 with low PSA density may safely avoid biopsy.
- **Transperineal biopsy** under local anaesthesia: near-zero sepsis; MRI–ultrasound fusion targeting. **Micro-ultrasound** (PRI-MUS scoring) matches mpMRI in registry data.
- **PSMA (prostate-specific membrane antigen) PET (positron emission tomography): proPSMA:** accuracy 92% vs 65% for CT (computed tomography) plus bone scan (sensitivity 85%, specificity 98%). Used for high-risk staging, biochemical recurrence and selecting radioligand therapy.

2. Localised disease

- **AS (active surveillance):** standard for GG1; extended to low-volume favourable intermediate risk (GG2), with MRI and PSA density. **ProtecT (15 years):** prostate cancer mortality about 2–3% in monitoring, surgery and radiotherapy arms; more metastasis on monitoring, no survival difference.
- **Robotic RP (radical prostatectomy): Retzius-sparing** (Bocciardi) gives earlier continence (1 week to 3 months; equal by 12 months); single-port robot; frozen-section-

guided nerve sparing (NeuroSAFE).

- **Radiotherapy:**
  - Moderate hypofractionation **60 Gy in 20** (CHHiP) is standard.
  - **SBRT (stereotactic body radiotherapy) 36.25 Gy in 5:** PACE-B showed similar late toxicity at 2 years (chapter); its 5-year report (2024) showed non-inferior biochemical or clinical control. HYPO-RT-PC (42.7 Gy in 7) was also non-inferior.
  - **FLAME:** MRI-lesion boost improved biochemical PFS (progression-free survival) (HR [hazard ratio] 0.45). **Hydrogel rectal spacer** reduces rectal toxicity.
  - High-risk M0: **abiraterone added to ADT (androgen deprivation therapy) and RT (STAMPEDE):** MFS (metastasis-free survival) HR 0.53.
- **Focal therapy:** based on the **index lesion hypothesis** (the largest or highest-grade focus drives progression). Energies: HIFU (high-intensity focused ultrasound), cryotherapy, IRE (irreversible electroporation), focal laser, VTP (vascular-targeted photodynamic therapy, padeliporfin). **HEAT registry (1379 men):** failure-free survival 68% (intermediate risk); continence 90–95%. Still within trials or registries.

3. Recurrent and advanced disease

Setting	Advance	Evidence
High-risk BCR (biochemical recurrence)	PSMA-guided salvage; enzalutamide + leuprolide	**EMBARK:** MFS HR 0.42
mHSPC (metastatic hormone-sensitive)	ADT + ARPI (androgen receptor pathway inhibitor)	LATITUDE, TITAN, ENZAMET, ARCHES
	Triplet (ADT + docetaxel + ARPI)	**PEACE-1, ARASENS**
	Prostate RT, low volume	STAMPEDE arm H
	MDT (metastasis-directed therapy)	STOMP, ORIOLE
nmCRPC (non-metastatic castration-resistant)	Apalutamide, enzalutamide, darolutamide	SPARTAN, PROSPER, ARAMIS
HRR (homologous recombination repair)-mutated mCRPC	**PARP (poly-ADP-ribose polymerase) inhibitors**	**PROfound:** olaparib imaging PFS 7.4 vs 3.6 months; PROpel, MAGNITUDE, TALAPRO-2
PSMA-positive mCRPC	**^{177}Lu-PSMA-617**	**VISION:** OS (overall survival) 15.3 vs 11.3 months (HR 0.62); TheraP: PSA response 65% vs 37% vs cabazitaxel

- **ADT alone is no longer standard** for mHSPC; every fit man gets intensification.
- **Germline and somatic testing** for all metastatic disease; **pembrolizumab** for MSI-high (microsatellite instability-high) tumours.
- **Relugolix:** oral GnRH (gonadotropin-releasing hormone) antagonist; HERO showed better sustained castration than leuprolide.
- **Also know:** radium-223 (ALSYMPCA, OS HR 0.70); PSMAfore (lutetium before chemotherapy); AI in MRI reading and pathology.

Diagram to draw: MRI-first pathway: raised PSA → mpMRI → PI-RADS ≥3 targeted transperineal biopsy; PI-RADS 1–2 with low PSA density → surveillance.

Key points to remember:

- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
- proPSMA: PSMA PET accuracy 92% vs 65%.
- ProtecT: no mortality difference at 15 years; AS is standard for GG1.
- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Q3a. Muscle assisted urethroplasty

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 4, Q3b)

Read as muscle-flap-assisted graft urethroplasty: a vascularised muscle flap used as bed or cover for a graft. "Muscle-sparing" bulbar urethroplasty is noted at the end.

Rationale

- A free graft (e.g. **BMG [buccal mucosal graft]**) has no blood supply. It survives by **imbibition** (about 48 h, nutrients from the bed), then **inosculation** (new microcirculation). So **"a graft lives on its bed"**.
- **Poor beds:** radiation, multiple failed repairs, fistula, pelvic-fracture scar, infection. Here a **muscle flap brings blood supply**, fills dead space and separates suture lines.

Muscle flaps used

Flap	Supply	Use
Gracilis (muscle-only, axial, multiple pedicles)	Medial circumflex femoral branch, proximal thigh	Irradiated posterior urethra, rectourethral fistula, complex perineal repair
Bulbospongiosus	Perineal branches	Covers a ventral graft; support in bulbar repair
Rectus abdominis (VRAM, vertical rectus abdominis myocutaneous)	Inferior epigastric	Large pelvic or perineal defects

Technique (gracilis-assisted BMG)

1. Lithotomy; perineal exposure; excise scar or fistula.
2. Close rectal side; urethra opened.
3. Gracilis harvested through a medial thigh incision; pedicle preserved; tunnelled to perineum.
4. Muscle interposed between rectum and urethra, or used as the graft bed.
5. **BMG quilted onto the muscle** as onlay; urethra closed over a catheter.
6. Catheter drainage; urethrogram before removal.

Results (Chapter 86)

- Gracilis with or without BMG for rectourethral fistula: **non-irradiated, all closed**, no later strictures. **Irradiated:** most closed in one stage, but more strictures and permanent bowel diversion.
- Muscle interposition is integral; BMG is invaluable for urethral closure.

Complications

- Thigh wound problems; graft contraction, restricture, fistula recurrence.

"Muscle-sparing" bulbar urethroplasty (Barbagli)

- The **bulbospongiosus muscle, central perineal tendon and perineal nerves** are left intact; dorsal BMG onlay through a lateral approach.
- Aim: less post-void dribbling and ejaculatory dysfunction.

Diagram to draw: gracilis flap tunnelled to the perineum, lying between rectum and urethra with BMG on it.

Key points to remember:

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Q3b. Silodosin.

Type: SAQ (10 marks)

Introduction

- **Uroselective α1-blocker**, the **most α1A-selective** agent: several hundredfold over α1B, about 50-fold over α1D.

Mechanism of action

- **α1A receptors** (about 70% of prostatic α1 receptors) dominate in the prostate, bladder neck and prostatic urethra. Noradrenaline on them causes the **dynamic component** of BOO (bladder outlet obstruction).
- Blockade relaxes this smooth muscle. Little α1B (vascular) blockade means few blood-pressure effects.

Pharmacology

- **Dose: 8 mg once daily with a meal**; no titration.
- **Renal:** 4 mg if CrCl (creatinine clearance) 30–50 mL/min; contraindicated below 30. Contraindicated in severe hepatic impairment.
- Metabolised by CYP3A4: avoid potent CYP3A4 inhibitors (ketoconazole) and other α-blockers.
- Onset: flow within 2–6 h, symptoms within 3–4 days.

Indications

Indication	Evidence
LUTS (lower urinary tract symptoms) due to BPE (benign prostatic enlargement)	Japanese, American and European 12-week RCTs (randomised controlled trials): IPSS (International Prostate Symptom Score) and flow better than placebo; **non-inferior to tamsulosin 0.4 mg**; effective in severe LUTS
MET (medical expulsive therapy)	Distal ureteric stones 5–10 mm; meta-analyses show higher, faster expulsion than tamsulosin
CP/CPPS (chronic prostatitis/chronic pelvic pain syndrome)	α-blocker-naive men with voiding symptoms

- Greatest fall in BOOI (BOO index) of all α1-blockers (subgroup data). **Also know:** nocturia (modest).

Adverse effects

- **Ejaculatory dysfunction (signature):** anejaculation in over a quarter of men in registration trials (about 14% vs 2% with tamsulosin head-to-head). Mechanism: **emission failure** (vas and seminal vesicle α1A blockade) and bladder-neck relaxation. Reversible; few stop the drug. Counsel young men.
- **Cardiovascular:** dizziness; orthostatic hypotension rare; no QT change; safe with antihypertensives.
- **IFIS (intraoperative floppy iris syndrome):** class effect; tell the ophthalmologist before cataract surgery (stopping the drug does not remove risk).
- Falls in older men.

Comparison

	Tamsulosin	Silodosin
Selectivity	α1A and α1D	α1A ≫ α1D, α1B
Efficacy in LUTS	Equal	Equal
Ejaculatory disorder	Less	More

Diagram to draw: prostate and bladder neck showing α1A receptor sites blocked by silodosin.

> **Key points to remember:**
>
> - Most α1A-selective α-blocker; 8 mg with food.
> - Equal efficacy to tamsulosin 0.4 mg; more anejaculation.
> - Halve to 4 mg at CrCl 30–50; avoid below 30.

Q3c. Optical coherence tomography in carcinoma bladder.

Type: SAQ (10 marks)

Introduction

- **WLC (white-light cystoscopy)** is the gold standard but subjective: many biopsies are negative and small lesions are missed.
- **OCT (optical coherence tomography)** is an **"optical biopsy"** technique listed among enhanced cystoscopic methods for NMIBC (non-muscle-invasive bladder cancer).

Principle

- OCT is the **optical analogue of B-mode ultrasound**. It measures the echo delay of back-scattered **near-infrared light** instead of sound, using **low-coherence interferometry**.
- It gives real-time cross-sectional images with about 10–20 μm resolution, but only **1–2 mm depth**. First described in 1991 (ophthalmology).

Technique

- A fibre-optic probe passes through the cystoscope's working channel.
- WLC or blue light finds the lesion; the probe is placed **in contact** with it and read at once.

Image interpretation

Tissue	OCT appearance
Normal	Three layers: thin low-scatter urothelium, bright lamina propria, darker muscularis
Inflammation	Layers preserved but thickened
Papillary tumour (Ta)	Thickened urothelium, papillary pattern; lamina propria boundary intact
Invasive (T1)	**Loss of the urothelium–lamina propria boundary**

Clinical uses

- Separate benign from malignant flat lesions; fewer biopsies.
- Estimate Ta versus T1 during TURBT (transurethral resection of bladder tumour).
- Check resection margins and completeness.
- **UTUC (upper tract urothelial carcinoma):** through the ureteroscope, OCT correlates well with final grade and stage.

Evidence and limitations

- Small studies show **high sensitivity but lower specificity**: inflammation, scar and post-BCG (bacille Calmette–Guérin) change mimic tumour. Adding OCT to blue light improves specificity.
- Limitations: tiny field of view; too shallow to stage muscle invasion; no cellular detail, so cannot grade; cost; learning curve.
- Not recommended routinely by guidelines; investigational.

Other enhanced techniques

Technique	Principle
Blue-light (HAL, hexaminolevulinate)	Porphyrins fluoresce red; detects more CIS (carcinoma in situ)
NBI (narrow-band imaging)	415/540 nm light absorbed by haemoglobin
CLE (confocal laser endomicroscopy)	Cellular-level images after fluorescein

Diagram to draw: OCT image showing three bladder layers, and a tumour with loss of layering.

> **Key points to remember:**
>
> - Near-infrared "ultrasound with light"; depth 1–2 mm.
> - Loss of layering suggests tumour.
> - High sensitivity, false positives from inflammation; investigational.

Q3d. Re-look TUR bladder tumour.

Type: SAQ (10 marks)

Read as second-look (repeat or restaging) TURBT.

Definition

- A **planned second TURBT (transurethral resection of bladder tumour)** of the primary site, including detrusor muscle. It completes resection and corrects staging.

Rationale

- **TURBT is a staging operation first.** Completeness and muscle in the specimen are the quality markers.
- Residual tumour is common:
  - At the resection site in **≥40%** when re-resected within weeks.
  - **HG (high-grade) Ta:** residual up to 50%, upstaging up to 15%.
  - **HG T1:** upstaging to muscle invasion in **25–30%**, up to about half if **no muscle** in the first specimen.
  - Understaging is about twice as likely without muscle.

Indications

Recommended	Consider	Not needed
Incomplete initial resection	HG Ta, multifocal or large (AUA [American Urological Association])	Low-grade Ta
All T1 tumours	Outside resection with no slides for review	Primary CIS (carcinoma in situ)
No detrusor muscle in specimen (except low-grade Ta and primary CIS)		Cystectomy already planned

Timing

- **2–6 weeks** after the first TURBT (EAU [European Association of Urology]); **within 6 weeks** (AUA). This allows healing yet avoids delaying intravesical BCG (bacille Calmette–Guérin).

Technique

- Resect the **primary-site scar deep into detrusor**, plus any residual tumour; send specimens separately. Blue-light or NBI (narrow-band imaging) helps.

Evidence

- Changes treatment in about **one-third**.
- **Divrik (2010) RCT (randomised controlled trial)** in T1: lower recurrence and progression.
- Better BCG response when BCG follows re-TUR.
- **Residual T1 at re-resection** predicts progression in up to **80%**: consider early cystectomy.

Decision after re-TUR

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No tumour / Ta            → BCG induction + maintenance
Residual HG T1            → early cystectomy (or BCG with close review)
Muscle invasion (≥T2)     → neoadjuvant chemotherapy + cystectomy
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Recent advances

- **En bloc resection:** more detrusor, less residual tumour.
- Expert pathology review changes management in about one-third.
- **Also know:** maximal TURBT before trimodality therapy.

Diagram to draw: bladder wall showing first resection scar and deep second resection into detrusor.

> **Key points to remember:**
> - Indications: incomplete resection, all T1, no muscle.
> - Timing 2–6 weeks.
> - HG T1 upstaged in 25–30% (about half without muscle).
> - Residual T1 progresses in up to 80%.

Q3e. STING technique for VUR.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3c)

Definition and principle

- **STING (subureteric Teflon injection)**: endoscopic injection of a **bulking agent** beneath the ureteric orifice in VUR (vesicoureteral reflux). The mound supports the intramural ureter, restoring the **flap-valve**.
- Matouschek, 1981; popularised by **O'Donnell and Puri, 1986**.

Bulking agents

Agent	Particle	Problem
Teflon (PTFE (polytetrafluoroethylene) paste)	90% <40 µm, permanent	**Migration**; never FDA (Food and Drug Administration) approved
Dx/HA (dextranomer/hyaluronic acid, Deflux)	**80–250 µm**; gel resorbed	Volume loss about 23%; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Late ureteric obstruction

Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING**: needle enters **2–3 mm below the orifice at 6 o'clock**, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.
4. Aim for a **"volcano" mound** with a slit-like orifice. Day case.

Technique	Needle entry	Result
STING	Below orifice	Baseline
HIT (hydrodistension implantation technique)	Inside hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Indications

- Persistent primary VUR (grades II–IV) with breakthrough febrile UTI (urinary tract infection), new scars or parental preference; residual VUR after reimplantation.
- **Relative contraindications**: grade V, **UVJ (ureterovesical junction) obstruction**, untreated BBD (bladder and bowel dysfunction).

Results

- Low-grade VUR about 90% after one injection. **Elder 2006 meta-analysis**: grade I–II about 79%, III 72%, IV 63%, V 51%.
- **Swedish Reflux Trial**: 20% recurrence at 2 years.

Complications

- Failure or recurrence (repeat 1–2 injections, then reimplant); new contralateral VUR; **early or late obstruction**; calcified mound mimics a stone. **Also know:** Macroplastique.

Follow-up

- Ultrasound at 2–6 weeks; ultrasound ± VCUG (voiding cystourethrogram) at 3 months.

Diagram to draw: needle sites: STING below orifice, HIT and double HIT inside the tunnel, with volcano mound.

> **Key points to remember:**
>
> - O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
> - Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
> - Watch late obstruction; calcified mound mimics stone.

Q3f. Diet in stone disease.

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 4, Q3f)

Principle

- Diet lowers urinary **supersaturation** by **dilution**, by reducing promoters (calcium, oxalate, uric acid, sodium) and by raising inhibitors (citrate).
- General advice for all; recurrent formers tailored to **24-hour urine**.

General measures

Measure	Target	Mechanism and evidence
Fluid	Urine **≥2.5 L/day** (about 3 L intake), spread through the day	Most important single factor. **Borghi 1996 RCT (randomised controlled trial):** recurrence 12.1% vs 27% at 5 years
Normal calcium	1000–1200 mg/day, with meals	Binds oxalate in the gut. **Restriction increases stones**. **Borghi 2002:** normal-calcium, low-salt, low-protein diet halved stone events vs low-calcium diet
Low sodium	Salt about 4–5 g/day	Sodium increases urinary calcium, lowers citrate
Limit animal protein	0.8–1.0 g/kg/day	Acid and purine load: hypercalciuria, hypocitraturia, hyperuricosuria
Fruit and vegetables (DASH [Dietary Approaches to Stop Hypertension] diet)	Daily	Alkali and citrate; DASH linked to fewer stones
Weight	BMI (body mass index) 18–25	Obesity: low urine pH, more oxalate and uric acid

- **Citrus:** lemonade (most evidence) and orange juice raise citrate; orange juice adds alkali. **Volume matters more than drink type.**
- Avoid sugar-sweetened drinks and **high-dose vitamin C** (converted to oxalate).

Stone-specific diet

Stone or abnormality	Diet
Hyperoxaluria	Avoid large portions of **spinach, beets, chocolate, nuts, tea**; calcium with meals
Enteric hyperoxaluria	Strict low oxalate and **low fat**; **calcium up to 1 g/day with meals**; fluid; magnesium
Uric acid	Less animal protein (purines give about 30% of urinary uric acid); fruit and vegetables; weight loss
Cystine	Fluid **3–5 L/day** including night; sodium **≤2300 mg/day**; less animal protein (not in children)
Struvite	Infection control; diet has little role

Adherence

- Poor: only half reach 2.5 L at 6 months; apps and smart bottles help.

Diagram to draw: "stone-prevention plate": water, fruit and vegetables, normal dairy, little meat and salt.

Key points to remember:

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.

Rapid Revision Sheet – Practice Set 5 Paper 4

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe energy sources for braking stones in uro (LAQ)

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- PROMIS and PRECISION: MRI before biopsy finds more significant cancer and avoids biopsies.
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- VISION (Lu-PSMA, HR 0.62); PEACE-1 and ARASENS triplets; EMBARK.

Q3a. Muscle assisted urethroplasty (SAQ)

- Graft survival: imbibition then inosculation.
- Muscle flap rescues a poor (irradiated) bed.
- Gracilis is axial, robust, easy to tunnel.
- Muscle-sparing: preserves bulbospongiosus and nerves.

Q3b. Silodosin. (SAQ)

- Most α1A-selective α-blocker; 8 mg with food.
- Equal efficacy to tamsulosin 0.4 mg; more anejaculation.
- Halve to 4 mg at CrCl 30–50; avoid below 30.

Q3c. Optical coherence tomography in carcinoma bladder. (SAQ)

- Near-infrared "ultrasound with light"; depth 1–2 mm.
- Loss of layering suggests tumour.
- High sensitivity, false positives from inflammation; investigational.

Q3d. Re-look TUR bladder tumour. (SAQ)

- Indications: incomplete resection, all T1, no muscle.
- Timing 2–6 weeks.
- HG T1 upstaged in 25–30% (about half without muscle).
- Residual T1 progresses in up to 80%.

Q3e. STING technique for VUR. (SAQ)

- O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%.
- Watch late obstruction; calcified mound mimics stone.

Q3f. Diet in stone disease. (SAQ)

- Urine ≥2.5 L/day (Borghi 1996).
- Never restrict calcium (Borghi 2002).
- Low salt, less animal protein, more fruit and vegetables.
- Cystine: ≥3 L urine; enteric: low fat, calcium with meals.